Applied Crypto

Exploring Block Cipher and the Meet in the Middle Attack

Problem 1 Explore different modes of operation through manual encryption and decryption.

Objective: Gain practical insights into the modes of operation discussed in class by manually encrypting and decrypting a given plaintext.

Task Details:

1. Encryption Setup:

a) Use a hypothetical block cipher with a block length of 4, defined as $E_k(b_1b_2b_3b_4) = (b_2b_3b_1b_4)$.

b) Convert English plaintext into a bit string using the table provided (A=0000 to P=1111). Assume we have a language that uses 16 letters only. If we want a more realistic exercise, we can have block size of 5 bits that can represent 32 cases (more than 26 letters) or 8 bits that use the ASCII.

A	B	C	D	E	F	G	H
0000	0001	0010	0011	0100	0101	0110	0111
I	J	K	L	M	N	O	P
1000	1001	1010	1011	1100	1101	1110	1111

2. Encryption Modes: Encrypt the plaintext 'FOO' using the following modes. Convert the final ciphertexts into letters. Show your work

a) ECB (Electronic Codebook)

b) CBC (Cipher Block Chaining) with IV=1010

c) CTR (Counter) with ctr=1010

3. Decryption Task:

Assume the ciphertexts from 2 are received by an intended receiver.

a) Manually decrypt each ciphertext to recover the original plaintext. Show your work.

Problem 2 Implementing a Meet-in-the-Middle (MITM) Attack on a Mini Block Cipher

Project Overview:

This project involves students in the implementation and analysis of a meet-in-the-middle (MITM) attack against a simplified block cipher called mini block cipher based on the SAES[1]. The primary goal is to provide hands-on experience with the MITM attack, showcasing its effectiveness against certain

[1] Refer to the detail implementation inside the lecture notes on Block Ciphers

cryptographic algorithms and understanding why modern ciphers like AES are designed to be immune to such attacks.

The MITM attack is an exhaustive key search attack[2]. It is a cryptanalytic attack applicable to ciphers based on composition of multiple rounds of substitutions and permutations. It works by finding plaintext-ciphertext pairs that map to the same intermediate value after partial encryption/decryption[3].

We designed a mini-block cipher based on SAES that starts with the key expansion, then works on encryption to get ciphertext and on decryption to recover the plaintext. The key expansion generates two keys, K1 and K2. K1 is used to perform Round 1 transformation on state, defined as encrypt_round1(). K2 is used to perform Round 2 transformations on state, defined as encrypt_round2(). For decryption, it is reverse. K2 is used to perform inverse Round 2 transformations on ciphertext, defined as decrypt_round2(). K1 is used to perform inverse Round 1 transformations on state, defined as decrypt_round1(). So, the pseudo-code for this block cipher would be:

1. Get (Key1, Key2) from Key K using the key expansion.
2. Two rounds of Encrypt to get the ciphertext.
    a. encrypt_round1() using K1
        i. Subsititute()
        ii. Shift()
        iii. Mix()
        iv. AddRoundKey()
    b. encrypt_round2() using K2
        i. Subsititute()
        ii. Shift()
        iii. AddRoundKey()

3. Two rounds of decryption to recover the plaintext
    a. decrypt_round2() using K2
        i. AddRoundKey()
        ii. Shift()
        iii. Subsititute()
    b. decrypt_round1() using K1
        i. AddRoundKey()
        ii. Mix()
        iii. Shift()
        iv. Subsititute()

Note, the only difference of this mini block cipher to the SAES is that it does not perform the first addroundkey() with the original K.

The MITM attack strategy:

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[2] https://en.wikipedia.org/wiki/Meet-in-the-middle_attack
[3] For example, https://youtu.be/S-EhbhDXUwM

a. Calculate X = encrypt_round1(K1, P)
b. Calculate X' = decrypt_round2(K2, C).
c. Find out a pair (K1, K2) such at X = X'
d. For one specific (P, C), there might be many matched pairs, we need to use more plaintext and ciphertext pairs to eliminate some of the matched pairs. Ideally, we shall get one matched key pair.
e. We can recover the plaintext from any ciphertext by using the key pair. In other words, we cracked the block cipher.

Project Tasks

Task 1: Implementing Mini Block Cipher with key size 16 bit and block size 16 bit:

a) Students will implement the Mini Block Cipher encryption and decryption functions using jupyter notebook. That is, students need to implement the above pseudo-code

b) Make at least ten pairs of plaintexts and ciphertexts using the above implementation.

Task 2: MITM Attack Implementation:

a) Students need to implement the meet in the middle attack strategy to mini block cipher, that is the strategy a-d.

b) Show the results.

Task 3: Analyze the time and memory complexity of the attack compared with the naive exhaustive key search.

a) What is the key space for the mini block cipher?
b) Image the mini block cipher is executed twice to generate a cipher text. It is called double mini cipher block. We need a key in 32 bits. The first 16 to the first mini block cipher, the remaining 16 to the second mini block cipher. The meet in the middle attack is to match the state for the first encryption of mini block cipher and the second decryption mini block. How many operations are needed to such attack?
c) If we do exhaustive key search for the double mini block cipher, how many operations are needed?
d) What is the trade off in this attack?

Task 4: Discuss how we can improve the mini block cipher.

Students may start by analyzing whether the SAES is vulnerable to the MITM attack or not. Remember SAES is the mini block cipher except that it has the initial addoundkey() using the original key K.

Deliverables and Submission

- Written Report in pdf
  - Students shall merge the exported pdf from jupyter notebook with other work into one pdf.
- Code: Use Jupyter Notebook

Grading rubrics

	Max Point	Expectations from the description. Show your work
Problem 1	**2.5**	
2a	0.5	Accurately perform ECB encryption for 'FOO', converting ciphertext into letters.
2b	0.5	Accurately perform CBC with IV=1010 encryption for 'FOO', converting ciphertext into letters.
2c	0.5	Accurately perform CTR with IV=1010 encryption for 'FOO', converting ciphertext into letters.
3a	1.0	Successfully decrypts each ciphertext, demonstrating understanding of decryption processes and converting plaintext back into letters.
Problem 2	**7.5**	
Task 1a	2.5	Accurate implementation of Mini Block Cipher functions in Jupyter Notebook, following provided specifications.
Task 1b	0.5	Creation of at least ten distinct plaintext-ciphertext pairs, clearly presented.
Task 2a	1.5	Correct implementation of meet-in-the-middle attack strategy on the mini block cipher, demonstrating understanding.
Task 2b	1	Clear presentation of meet-in-the-middle attack results using plaintext-ciphertext pairs from Task 1b.
Task 3a	0.3	Correct Calculation of key space for the mini block cipher.
Task 3b	0.3	Correct calculation of operations required for the meet-in-the-middle attack on the double mini block cipher
Task 3c	0.4	Correct calculation of operations needed for exhaustive key search on the double mini block cipher.
Task 3d	0.5	Clear discussion of trade-offs between meet-in-the-middle attack and exhaustive key search, demonstrating critical thinking.
Task 4	0.5	List couple of improvements to mitigate the MITM attack.